

Bilal Shihab

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PROFESSIONAL SUMMARY

Biomedical Engineering undergraduate specializing in AI, Cloud, and Embedded Software. Experienced in developing inference pipelines and optimizing machine learning models for edge devices. Passionate about leveraging Cloud and Edge AI to solve complex biological data challenges.

EDUCATION

UNIVERSITY OF TEXAS AT AUSTIN

Austin, TX

Bachelor of Science in Biomedical Engineering, Computational Track

May 2027

- **Relevant Coursework:** Embedded Systems, Biomedical Instrumentation, Circuits, Systems & Signals, Differential Equations & Linear Algebra, Intro to Computational Engineering Design, Statistics, Numerical Methods

SKILLS

- **Languages:** Python, C, ARMv6-M Assembly, MATLAB, R
- **AI & Data:** PyTorch, TensorFlow Lite, Google Cloud Platform (GCP), Optuna, Captum, Docker
- **Embedded & Hardware:** TI MSPM0, Raspberry Pi, Arduino, Code Composer Studio, UART, SolidWorks

EXPERIENCE

LONGHORN NEUROTECH

Austin, TX

Edge AI Team Lead

Sep 2025 - Present

- Directed a team of student engineers to optimize neural networks for deployment on embedded systems (Raspberry Pi series).
- Designing a reusable quantization pipeline using ExecuTorch and an inference backend to support keyword spotting and future neuroprosthetic control applications.
- Developed an optimized ONNX inference pipeline to process EMG band signals for prosthetic arm control, reducing MLP model energy consumption by 45%.

AI/ML Developer

Sep 2024 - Aug 2025

- Developed machine learning models for Brain-Computer Interfaces (BCI) aimed at controlling a prosthetic arm, using architectures such as CNNs and Capsule Networks.
- Improved BCI model accuracy from 60% to 80% by tuning hyperparameters with Optuna and using Captum to reduce overfitting.

WANG LABORATORY, UT AUSTIN

Austin, TX

Machine Learning Researcher

Feb 2026 - Present

- Gut-Brain Axis Research: Writing machine learning models to analyze EEG and EGG data, predicting the optimal timing for targeted ultrasound vagus nerve stimulation to help alleviate gastrointestinal issues.
- Neurawear Startup Initiative: Collaborating with a student engineering team to develop the "Neustim" wearable device, processing epilepsy datasets to build predictive neuromodulation algorithms.

FUNCTIONAL OPTICAL IMAGING LAB, UT AUSTIN

Austin, TX

Researcher and Programmer

Sep 2023 - May 2025

- Developed a real-time Laser Speckle UI in Python, using Basler cameras and Arduinos for automated laser intensity control.
- Enhanced speckle pattern evaluation by implementing contrast analysis algorithms to maximize measurement precision.
- Refactored contrast adjustment scripts from MATLAB to Python, enabling integration with the real-time Laser Speckle Analysis UI.

TECHNICAL PROJECTS

MEDADVISOR - CLINICAL FEEDBACK APP

Jun 2026 - Present

- Built an iOS app with a Stanford surgeon that records medical consultations and scores them against a clinical rubric on-device.
- Deployed a 4-bit 7B LLM on iPhone (Llama.cpp, Metal GPU): full rubric analysis in under 4 minutes, running the 4 GB model in a 495 MB footprint via memory-mapped weights.
- Cut rubric evaluation latency ~60% (18.6 s → 7.3 s/criterion) by caching transcript KV state across all 16 evaluations; built an on-device benchmark that quantified 31% thermal throttling under sustained load.

CLOUD TO EDGE HEART RATE MONITOR

Dec 2025 - Present

- Architected a cloud-to-Raspberry Pi 5 data pipeline streaming real-time heart rate data for patient simulation.
- Developed a UART communication protocol to stream simulated patient data from the Pi 5 to a Pi Pico for low-latency processing.
- Built a custom C-based inference engine on the Pico's dual-core Cortex-M0+ architecture to execute quantized neural networks on the incoming data stream in order to classify the patient's heart condition.

EMBEDDED MULTIPLAYER RACER (ARM CORTEX-M0+)

Nov 2025 - Dec 2025

- Earned 'Best Embedded Design' in a class-wide competition for excellence in hardware-software integration.
- Built a bare-metal multiplayer racing game in C for a TI MSPM0 microcontroller, implementing collision detection, velocity-based physics, and sprite rendering.
- Programmed register-level drivers for UART, ADC, and DAC to interface with analog joysticks and audio peripherals, bypassing standard libraries for efficiency.
- Designed a custom UART protocol to synchronize player coordinates and game events between two microcontrollers in real-time.